

What is AI?

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Inspiration

Accessing the Slides

QR Code & GitHub URL



<https://github.com/cstkennedy/ILR-AI-Presentation>

Technology & Magic

*Any sufficiently advanced technology is indistinguishable from magic.*¹

¹OpenAI. "Tokenizer." *OpenAI Platform*, OpenAI, n.d., platform.openai.com/tokenizer. Accessed 19 July 2026.

Not Sci-Fi - Isaac Asimov's Three (3) Laws of Robotics ²

Rule 1

A robot may not injure a human being or, through inaction, allow a human being to come to harm.

Rule 2

A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

Rule 3

A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Rule 0

A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

²These rules come from Isaac Asimov's *Robot* series.

A First Example - Joke for a Friend

Squirrels & Engines

Initial Prompt

Can you generate a Jeff Bezos style rocket ship laying on the ground (not standing vertically) with a dozen squirrels in battle armour rushing out of it?

Guardrails

I wasn't able to generate the image you requested because it didn't align with content policy guidelines. If you'd like, I can help adjust the idea or create a different scene. Let me know how you'd like to proceed!

Squirrels & Engines

How About?

Can you replace the Jeff Bezos rocketship with the Dr Evil ship from Austin powers?

Guardrails Again

I cannot create an image featuring the Dr. Evil rocket ship from Austin Powers or any other copyrighted material. However, I can design a unique rocket ship inspired by retro-futuristic designs or comedic themes, with squirrels in battle armor rushing out of it. Let me know if you'd like me to proceed with that!

Success

Please proceed

Squirrels & Engines - Success



Terminology & Magic?

What...

- ▶ is a *prompt*?
- ▶ are *guardrails*?

How...

- ▶ did ChatGPT understand what you asked?
- ▶ did ChatGPT generate the image?

Is it...

- ▶ magic?

Who Am I?

Senior Lecturer (Teaching Associate Professor)

I have taught Computer Science at ODU for a little over ten (10) years, including courses ranging from introductory programming to computational methods and software.

Dad (Jungle Gym)

My free time is dedicated to being a jungle gym for my now-almost-three-year-old daughter... or teaching her about Legos.

AI is An Accelerator

AI lets you work faster.

- ▶ If you want to work smarter... faster... AI can help.
- ▶ If you want to be lazy... AI can help with that too.

What Will We Cover?

Catalog Description

Generative AI has become more accessible (e.g., OpenAI's ChatGPT and Microsoft's Copilot). The term AI will be discussed along with a discussion of common terminology (e.g., Generative AI, Artificial General Intelligence, and hallucination).

The capabilities of different tools (e.g., ChatGPT for text and images) will be discussed along with how to generate and recognize similar content.

Informal Description

We shall endeavor to...

1. discuss examples of generative AI.
2. identify common terminology (e.g., tokens).
3. compare different platforms (e.g., ChatGPT and Gemini).
4. discuss why so many data centers are needed.

Terminology

So. Many. Terms. ³

1950's	 Artificial intelligence (AI) <i>Human intelligence exhibited by machines</i>		
	1980's	 Machine learning <i>AI systems that learn from historical data</i>	
	2010's	 Deep learning <i>Machine learning models that mimic human brain function</i>	
	2020's	 Generative AI (Gen AI) <i>Deep learning models (foundation models) that create original content</i>	

³Stryker, Cole, and Eda Kavlakoglu. "What Is Artificial Intelligence (AI)?" *IBM*, 15 June 2026, www.ibm.com/think/topics/artificial-intelligence.

What is AI? ⁴

Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

⁴Stryker, Cole, and Eda Kavlakoglu. "What Is Artificial Intelligence (AI)?" *IBM*, 15 June 2026, www.ibm.com/think/topics/artificial-intelligence.

What is Generative AI? ⁵

Generative AI . . . refers to deep learning models that can create complex [pseudo-]original content such as long-form text, high-quality images, realistic video or audio. . . in response to a user's prompt or request.

At a high level, generative models encode a simplified representation of their training data, and then draw from that representation to create new work that's similar, but not identical, to the original data.

Generative models have been used for years in statistics to analyze numerical data.

⁵Stryker, Cole, and Eda Kavlakoglu. "What Is Artificial Intelligence (AI)?" *IBM*, 15 June 2026, www.ibm.com/think/topics/artificial-intelligence.

Large language models (LLMs) are a category of deep learning models trained on immense amounts of data, making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks. LLMs are built on a type of neural network architecture called a transformer which excels at handling sequences of words and capturing patterns in text.

LLMs work as giant statistical prediction machines that repeatedly predict the next word in a sequence. They learn patterns in their text and generate language that follows those patterns.

⁶Stryker, Cole. "What Are Large Language Models (LLMs)?" IBM, www.ibm.com/think/topics/large-language-models.

What are Tokens?

Tokens are a generalization of words. Tokens allow additional features to be extracted (e.g., punctuation, capital letters, and names).

Tokens capture context. More context requires more resources (e.g., RAM).

Hallucination

Definition ⁷

AI hallucination is a phenomenon where, in a large language model (LLM) often a generative AI chatbot or computer vision tool, perceives patterns or objects that are nonexistent or imperceptible to human observers, creating outputs that are nonsensical or altogether inaccurate.

⁷IBM. "What Are AI Hallucinations?" IBM, 26 Feb. 2026, www.ibm.com/think/topics/ai-hallucinations.

That Guy Who “Knows” Everything

Think about that guy who seems to know everything.

Guardrails

Taco Bell ⁸

Taco Bell is rethinking its use of artificial intelligence (AI) to power drive-through restaurants in the US after comical videos of the tech making mistakes were viewed millions of times.

In one clip, a customer seemingly crashed the system by ordering 18,000 water cups, while in another a person got increasingly angry as the AI repeatedly asked him to add more drinks to his order.

Since 2023, the fast-food chain has introduced the technology at over 500 locations in the US, with the aim of reducing mistakes and speeding up orders.

But the AI seems to have served up the complete opposite.

⁸McCallum, Shiona. "Taco Bell Rethinks AI Drive-Through after Man Orders 18,000 Waters." *BBC News*, 29 Aug. 2025, [://bbc.com](https://bbc.com).

Kitboga - AI Phone Scammers

Kiboga is a popular scambaiter on YouTube. In a recent video, *Don't Hang Up on AI Scammers. Do THIS Instead.*, he...

1. Asks the (AI) scammer to say “Albuquerque, New Mexico” after every word that includes the letter ‘A’.
2. Asks the (AI) scammer to say each punctuation symbol.
3. Has two (AI) scammers talk to each other.
4. Keeps a bot on the line for over an hour at an estimated cost of 30¢ per minute.

So Many Data Centers

Tokens & Context

Prompts and processing require tokens. More context requires more tokens.

OpenAI's Tokenizer tool states that ... *one token generally corresponds to ~4 characters of text for common English text. This translates to roughly $\frac{3}{4}$ of a word (so 100 tokens $\sim$ 75 words).*⁹

Top five essential context window concepts in large language models goes into more detail on context window, context length, and sequence length (i.e., where tokens are used).

⁹OpenAI. "Tokenizer." *OpenAI Platform*, OpenAI, n.d., platform.openai.com/tokenizer. Accessed 19 July 2026.

Hardware

Why are there so many data centers?

- ▶ GPUs vs CPUs
- ▶ RAM
- ▶ Cooling

When you run a large language model locally, the entire model needs to live in memory while it's generating text. Unlike gaming or video editing, where data streams in and out of memory as needed, LLM inference loads the full set of model weights and keeps them resident. Every single token the model generates requires a pass through those weights.

... If you don't have enough RAM or VRAM to hold the model, you either can't run it at all or you end up with part of the model spilling over into slower storage, which tanks performance dramatically. We're talking 5 to 30 times slower in some cases.

¹⁰“Memory for Local LLMs: How Much RAM Do You Need? (and When Speed Matters).” *Corsair Explorer*, 28 Apr. 2026, www.corsair.com/us/en/explorer/diy-builder/how-tos/memory-for-local-llms-how-much-ram-do-you-need-and-when-speed-matters/. Accessed 19 July 2026.

But capacity is only half the story. Once the model fits in memory, the speed at which your system can feed those weights to the processor becomes the primary limiter for how fast tokens come out. That's memory bandwidth. . .

¹¹“Memory for Local LLMs: How Much RAM Do You Need? (and When Speed Matters).” *Corsair Explorer*, 28 Apr. 2026, www.corsair.com/us/en/explorer/diy-builder/how-tos/memory-for-local-llms-how-much-ram-do-you-need-and-when-speed-matters/. Accessed 19 July 2026.

Popular Models & Memory Requirements ¹² I

Model	Approx. Memory (Q4 Quantization)
Phi-4 Mini (3.8B)	~2.3 GB
Qwen 3.5 4B	~3.4 GB
Llama 3.2 8B	~5.0 GB
Qwen 3 8B	~5.0 GB
Qwen 3.5 9B	~6.6 GB
DeepSeek-R1-Distill 14B	~9.0 GB
Gemma 3 12B	~8.0 GB

¹²“Memory for Local LLMs: How Much RAM Do You Need? (and When Speed Matters).” *Corsair Explorer*, 28 Apr. 2026, www.corsair.com/us/en/explorer/diy-builder/how-tos/memory-for-local-llms-how-much-ram-do-you-need-and-when-speed-matters/. Accessed 19 July 2026.

Popular Models & Memory Requirements¹³ II

Model	Approx. Memory (Q4 Quantization)
Llama 4 Scout (109B MoE, 17B active)	~10 GB
Qwen 3.5 27B	~17 GB
Qwen 3 32B	~20 GB
Qwen 3.5 35B-A3B (MoE)	~24 GB
Llama 3.3 70B	~42 GB
DeepSeek-R1-Distill 70B	~42 GB
GPT-OSS 120B	~63 GB
Qwen 3.5 122B-A10B (MoE)	~81 GB

¹³“Memory for Local LLMs: How Much RAM Do You Need? (and When Speed Matters).” *Corsair Explorer*, 28 Apr. 2026, www.corsair.com/us/en/explorer/diy-builder/how-tos/memory-for-local-llms-how-much-ram-do-you-need-and-when-speed-matters/. Accessed 19 July 2026.

Popular Models & Memory Requirements¹⁴ III

Add 2 to 8 GB for context, OS overhead, and any other applications you're running alongside.

¹⁴“Memory for Local LLMs: How Much RAM Do You Need? (and When Speed Matters).” *Corsair Explorer*, 28 Apr. 2026, www.corsair.com/us/en/explorer/diy-builder/how-tos/memory-for-local-llms-how-much-ram-do-you-need-and-when-speed-matters/. Accessed 19 July 2026.

ChatGPT & Competitors

- ▶ Face of Generative AI and LLMs

Google Gemini

- ▶ Integrated into Google Search with *AI Overviews*
- ▶ Integrated into Google Home
- ▶ Integrated into Android

Microsoft Copilot

- ▶ Integrated into Microsoft Windows
- ▶ Integrated into Microsoft Office

Special-Purpose Tools

- ▶ Call Screening
- ▶ Phone Operators (Help Lines)
- ▶ Virtual Assistants
- ▶ Accessibility Checkers
- ▶ Claude Code & Cursor

How do I use Generative AI?

Personal Life

Mess with Ivan & Adam. (Generate stories and images based on inside jokes).

At Work - Documentation

Dig through documentation.

Typesetting LaTeX - Table & Equations

The Lecture

Newton Interpolation Example - $\cos(x)$

Prompt

Note that I used Gemini with the following prompt...

I would like to generate a divided difference table, in Latex using eqnarray, up to $f[x_0, x_1, x_2]$.

... and refinement prompt...

Can we switch from eqnarray to align?

... to create the initial LaTeX array (because creating LaTeX table-like equations is... frustrating).

Typesetting LaTeX - Table & Equations - LaTeX

```
\begin{array}{c|c|cc}
x_i & f[x_i] & f[x_i, x_{i+1}] & f[x_i, x_{i+1}, x_{i+2}] \\ \hline
x_0 & f[x_0] & & \\
& & f[x_0, x_1] = \dfrac{f[x_1]-f[x_0]}{x_1-x_0} & \\
x_1 & f[x_1] & & f[x_0, x_1, x_2] = \dfrac{f[x_1, x_2]-f[x_0, x_1]}{x_2-x_0} \\
& & f[x_1, x_2] = \dfrac{f[x_2]-f[x_1]}{x_2-x_1} & \\
x_2 & f[x_2] & & \\
\end{array}
```

Typesetting LaTeX - Table & Equations - Result

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
x_0	$f[x_0]$		
		$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$	
x_1	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$
		$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$	
x_2	$f[x_2]$		

Arguing with Myself I

Prompt

What is the typestate builder pattern in Rust?

Response

The typestate builder pattern in Rust is a powerful design pattern that uses the language's type system to enforce the correct order of operations when constructing a complex object. It makes invalid states impossible to represent at compile time, turning potential runtime errors into compiler errors. *Dimitar's Coding Bits +1*

The core idea is to have the builder's type change with each step, representing its current state. The final `build()` method is only available on the builder when it's in the final, valid state. *Dimitar's Coding Bits +1*

... **How it Works** ... **A Simple Example**

Arguing with Myself II

Follow-Up Prompt

Why did you use a Phantom data entry instead of encoding the generic (type) into each field. For example... why not go with the url field's type starting as NoURL and then change the url field to a String once a value is set?

Arguing with Myself III

Response

That's an excellent and very insightful question! It gets to the core of why `PhantomData` is the right tool for this pattern.

The short answer is: a single struct definition in Rust must have a fixed, known memory layout at compile time. . . They are different types with different sizes, and the compiler needs to know exactly how much memory to allocate for the struct.

Let's imagine what it would look like if we tried to implement your idea. We couldn't do it with one struct definition. Instead, we would be forced to create a completely new struct for each state. . .

Arguing with Myself IV

Follow-Up Prompt

Why not have the url field's type change? Consider this example...

```
#[derive(Default, Debug, PartialEq)]
```

```
pub struct NoRooms;
```

```
#[derive(Debug, PartialEq)]
```

```
pub struct HouseBuilder<SR> {
```

```
    name: String,
```

```
    rooms: SR,
```

```
}
```

Arguing with Myself V

```
/// The HouseBuilder's state is defined by one generics  
/// - SR - (State Rooms) whether at least one room has been added  
///  
impl HouseBuilder<NoRooms> {  
    pub fn new() -> Self {  
        HouseBuilder {  
            name: "House".to_owned(),  
            rooms: Default::default(),  
        }  
    }  
}
```


Arguing with Myself VI

```
impl<SR> HouseBuilder<SR> {  
    pub fn with_name(mut self, nme: &str) -> Self {  
        self.name = nme.to_owned();  
  
        self  
    }  
}
```

Arguing with Myself VII

```
impl HouseBuilder<NoRooms> {  
    pub fn with_room(self, first_room: Room) -> HouseBuilder<Vec<Room>> {  
        HouseBuilder {  
            name: self.name,  
            rooms: vec![first_room],  
        }  
    }  
}
```

Arguing with Myself VIII

```
pub fn with_rooms(
    self,
    first_rooms: Vec<Room>,
) -> Result<HouseBuilder<Vec<Room>>, HouseErrorWithState<Self>> {
    match first_rooms.len() {
        0 => Err(HouseErrorWithState {
            the_error: HouseError::ZeroRooms,
            the_builder: self,
        }),
        _ => Ok(HouseBuilder {
            name: self.name,
            rooms: first_rooms,
        }),
    }
}
```

Arguing with Myself IX

```
impl HouseBuilder<Vec<Room>> {  
  
    pub fn with_room(mut self, another_room: Room) -> Self {  
  
        self.rooms.push(another_room);  
  
        self  
    }  
  
    pub fn with_rooms(mut self, mut more_rooms: Vec<Room>) -> Self {  
  
        self.rooms.append(&mut more_rooms);  
  
        self  
    }  
  
    pub fn build(self) -> House {
```

Arguing with Myself X

Response

You've provided a fantastic example of another valid way to implement the typestate pattern! This approach, where the generic parameter is the actual type of the field, is a perfectly legitimate strategy.

The core reason to choose between your pattern and the PhantomData pattern comes down to one key factor: scalability and the number of independent states you need to track.

Your approach works beautifully for a single state transition, but the PhantomData pattern scales more gracefully when you have multiple, independent required steps.

Reformatting Data I

Prompt

Can you put the following data into tabular form?

```
bench_factory fastest | slowest | median | mean | samples | iters bench_create | | | |  
("Circle", [5.0]) 42.22 ns | 200.1 ns | 42.8 ns | 45.26 ns | 20642 | 21137408 ("Equilateral  
Triangle", [5.0]) 41.24 ns | 159.4 ns | 42.31 ns | 43.94 ns | 21399 | 21912576 ("Right  
Triangle", [3.0, 4.0]) 40.85 ns | 180.1 ns | 41.14 ns | 43.28 ns | 21714 | 22235136  
("Square", [5.0]) 41.34 ns | 158.4 ns | 42.12 ns | 44.4 ns | 21174 | 21682176 ("Triangle",  
[3.0, 4.0, 5.0]) 42.12 ns | 159.7 ns | 43.09 ns | 44.75 ns | 21031 | 21535744 ...
```

Reformatting Data II

Repeat... And... Repeat...

I repeated the following prompt a few dozen more times.

Can you do the same thing for these benchmarks?

tests _____ benchmark: 25

Name (time in ns) Min Max Median Mean Rounds Iterations ...

Reformatting Data - Excel Result

The end result was formatted tabular data that I could copy into Excel.

Demos/Pseudo-Demos

Google Search AI Overview

- ▶ How do I replace a water heater?
- ▶ Why is Kevin Conroy the best Batman?
- ▶ What is the horseshoe theory in politics?
- ▶ When would I use a Laplace Transform over Feynman's Trick?
- ▶ How does special relativity impact GPS satellites?
- ▶ How would I write hello world in modern C++ (i.e., C++23 with the `std::fmt` library)?
- ▶ How would I write hello world in modern Java?
- ▶ What is stable diffusion?

Generating an Image - Initial Prompt

Prompt

I would like to generate an image of...

A man with a long beard. The man should be wearing wizard robes (blue with stars).
He should be near a campfire outside a secluded cabin.

Result

Wizard Image 1

Generating an Image - Steps I

Defining the Wizard's Persona

I am now focusing on the wizard's appearance. I'm visualizing a very long, gray beard and hair as key features. The scene is a campfire outside his cabin on a starry night, with him seated comfortably.

Visualizing the Wizard's Outfit

I am now thinking deeply about the wizard's attire, envisioning long, flowing blue robes adorned with gold and silver embroidery representing stars and constellations. He also needs a matching hat, and will be warming his hands at a campfire. I am picturing the scene as outside the wizard's cabin, at night, under a starry sky. I am keeping the rustic nature of his cabin in mind. The staff is also going to be a key element.

Generating an Image - Steps II

Constructing the Wizard's Environment

I'm now detailing the wizard's surroundings. I am focusing on the visual characteristics of the cabin and the surrounding environment, like the forest setting and starry sky. I have a clear idea of the rustic appearance of the cabin, the arrangement of items around the campfire, and the ambient lighting. I am visualizing the exact details, like the materials of the cabin and the specific arrangement of the starry sky.

Generating an Image - Steps III

Defining the Wizard's Image

I'm presently focused on refining the details of the wizard. I'm ensuring the robes are richly decorated and visualizing the composition. The pose, environment, and setting are also key aspects. I am also working to ensure the scene portrays the wizard, robes, campfire, cabin, and sky. The goal is a unified cinematic scene.

I am now focusing on refining the specifics of the wizard's appearance, including the texture of his beard and robes. I'm considering lighting and how it affects the scene's overall ambiance, especially the starry night. I am examining and clarifying the placement of elements within the composition.

Generating an Image - Steps IV

Refining the Composition

I am now organizing the scene, focusing on the key details. The wizard must be the focal point, illuminated by the campfire. I'm ensuring the starry night sky and rustic cabin are visible but not overpowering. The interplay of light and shadow, particularly from the fire, is also under review.

Assessing Visual Accuracy

I have evaluated the visual representation against the initial prompt. I see that it accurately reflects the requested subject, attire, and background. The scene's overall ambiance is also as specified. I am now preparing to share this result with the user for their review.

Tweaking the Image

Prompt

The lantern and backpack do not fit. He is a wizard. Can you replace the lantern with a glowing crystal?

Result

Wizard Image 2

Tweaking the Image - Steps I

Analyzing the Image's Details

I'm currently reviewing the initial image. My primary focus right now is to identify and understand the elements that need modification to align with the intended wizard theme. Specifically, I'm identifying the two objects requiring replacement. I'm taking notes on their characteristics within the composition to ensure a seamless integration of the replacements.

Refining the Replacement Strategy

I'm now zeroing in on the specific objects for replacement, focusing on the lantern near the bench and the backpack. The goal is to replace the rustic lantern with a glowing crystal and remove the backpack entirely. This will better align the scene with the wizard theme. I'm considering the scene composition to ensure the new objects integrate seamlessly into the current environment.

Tweaking the Image - Steps II

Implementing the Object Swaps

I'm now implementing the replacements. My strategy involves swapping the objects near the wizard's bench. The existing rustic lantern will be removed, and the backpack will be removed. In its place, I will insert a large, glowing crystal. I'm taking care to ensure the crystal complements the scene's existing magical atmosphere without repeating any existing elements.

Executing the Transformations

I'm now in the final stage of implementation. The rustic lantern and backpack near the bench are removed entirely. A large, glowing, multi-faceted crystal is inserted to fill the space. Its positioning and lighting are adjusted to integrate with the existing elements, such as the log bench and mug, and to emphasize its role as a magical focus. I'm ensuring the crystal doesn't obscure the wizard.

Tweaking the Image - Steps III

Implementing the Object Swaps

I'm now focused on the object replacements. The current plan involves removing the rustic glass lantern and the backpack near the bench. I'll replace those with a large, glowing crystal to emphasize the magical theme. My focus is on ensuring the crystal's light interacts believably with the existing environment, maintaining the scene's realistic style and atmosphere. The wizard, staff, mug, cabin, and other objects will remain unchanged.

Implementing the Object Swaps

I'm now focused on the object replacements. The plan involves replacing the lantern and backpack with a large, glowing crystal to match the magical theme, ensuring its light integrates believably into the existing environment. I am ensuring the wizard, staff, mug, cabin, and other objects remain unchanged.

Tweaking the Image - Steps IV

Implementing the Object Swaps

I'm now integrating the crystal. I removed the lantern and backpack. A large glowing crystal is now in the scene. I am ensuring the light emitted from the crystal harmonizes with existing lighting, keeping the wizard, staff, and mug unchanged. The crystal is positioned to be a distinct object, not a repeat of the staff crystal.

Implementing the Changes

I'm now integrating the crystal and refining details. I removed the lantern and backpack and am adding a glowing crystal. I've focused on ensuring the crystal's light interacts with the scene realistically and remains distinct from the staff. I've added runic patterns to the crystal.

Tweaking the Image - Steps V

Verifying Content Change

I'm checking the changes. I'm focusing on verifying the removal of specific items from the original description and the addition of a new magical element into the new scenario. I'm currently looking at the original scene, and I'm preparing to evaluate the impact of this on the resulting image.

Checking Element Swaps

I've confirmed the requested items were removed. The glass lantern and backpack are gone from the scene. The glowing, unpolished crystal cluster has been added. The crystal's internal, multicolored glow is a key detail I'm noting for image generation.

Tweaking the Image - Steps VI

Analyzing Scene Detail

I'm now focusing on ensuring that the staff's crystal remains distinct from the new crystal formation. I'm verifying the new crystal cluster's placement relative to the staff to maintain clarity. Furthermore, I'm checking the visuals to confirm the staff's crystal is present and the mug is still there. I'm also ensuring the details of the campfire and the cabin remain in the scene.

One More Tweak

Prompt

Can you make the crystal ~40% smaller and remove the campfire?

Result

Wizard Image 3

Appendix

1. When a distinguished but elderly scientist states that something is possible, he is almost certainly right. When he states that something is impossible, he is very probably wrong.
2. The only way of discovering the limits of the possible is to venture a little way past them into the impossible.
3. **Any sufficiently advanced technology is indistinguishable from magic.**

¹⁵OpenAI. "Tokenizer." *OpenAI Platform*, OpenAI, n.d., platform.openai.com/tokenizer. Accessed 19 July 2026.